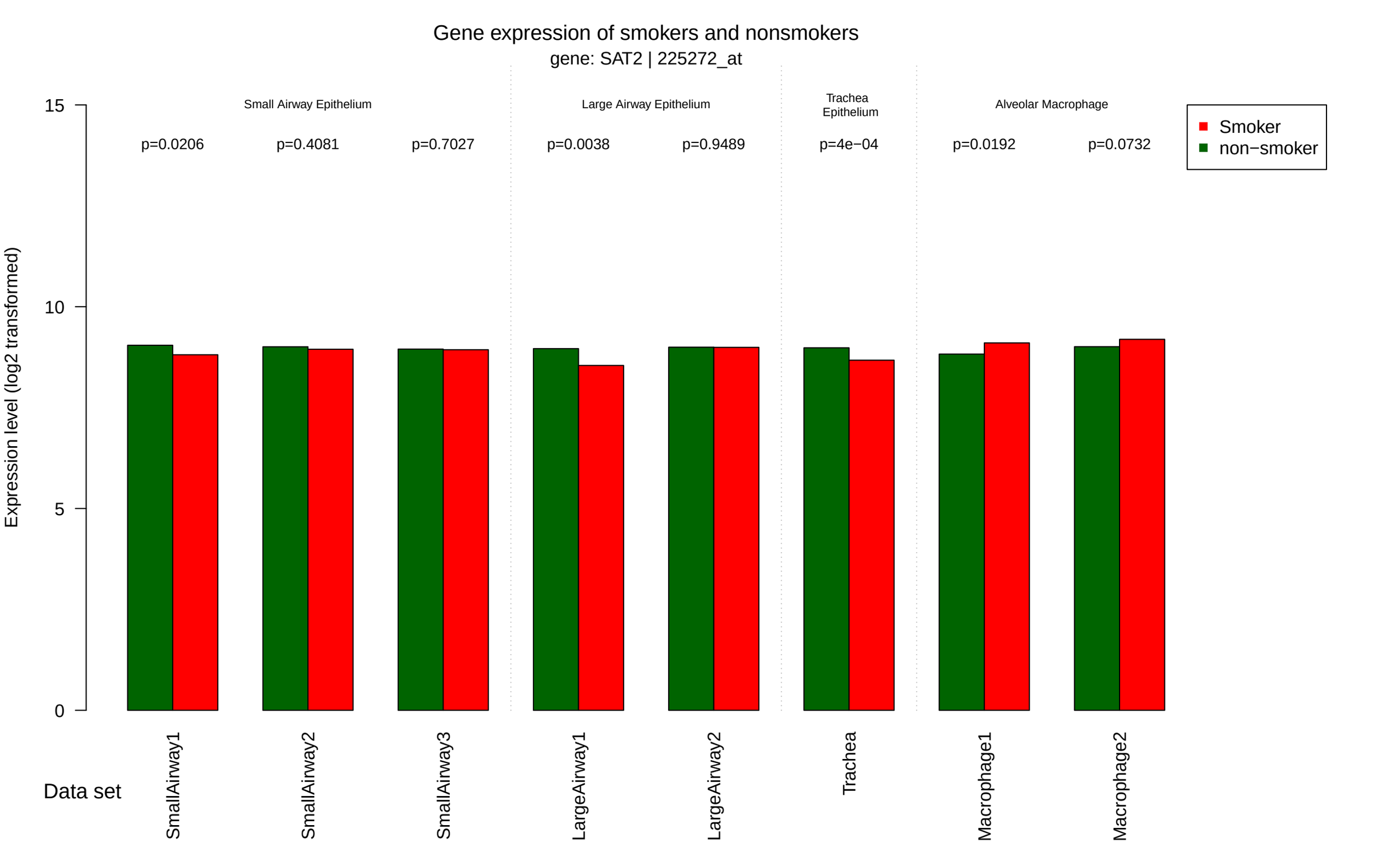
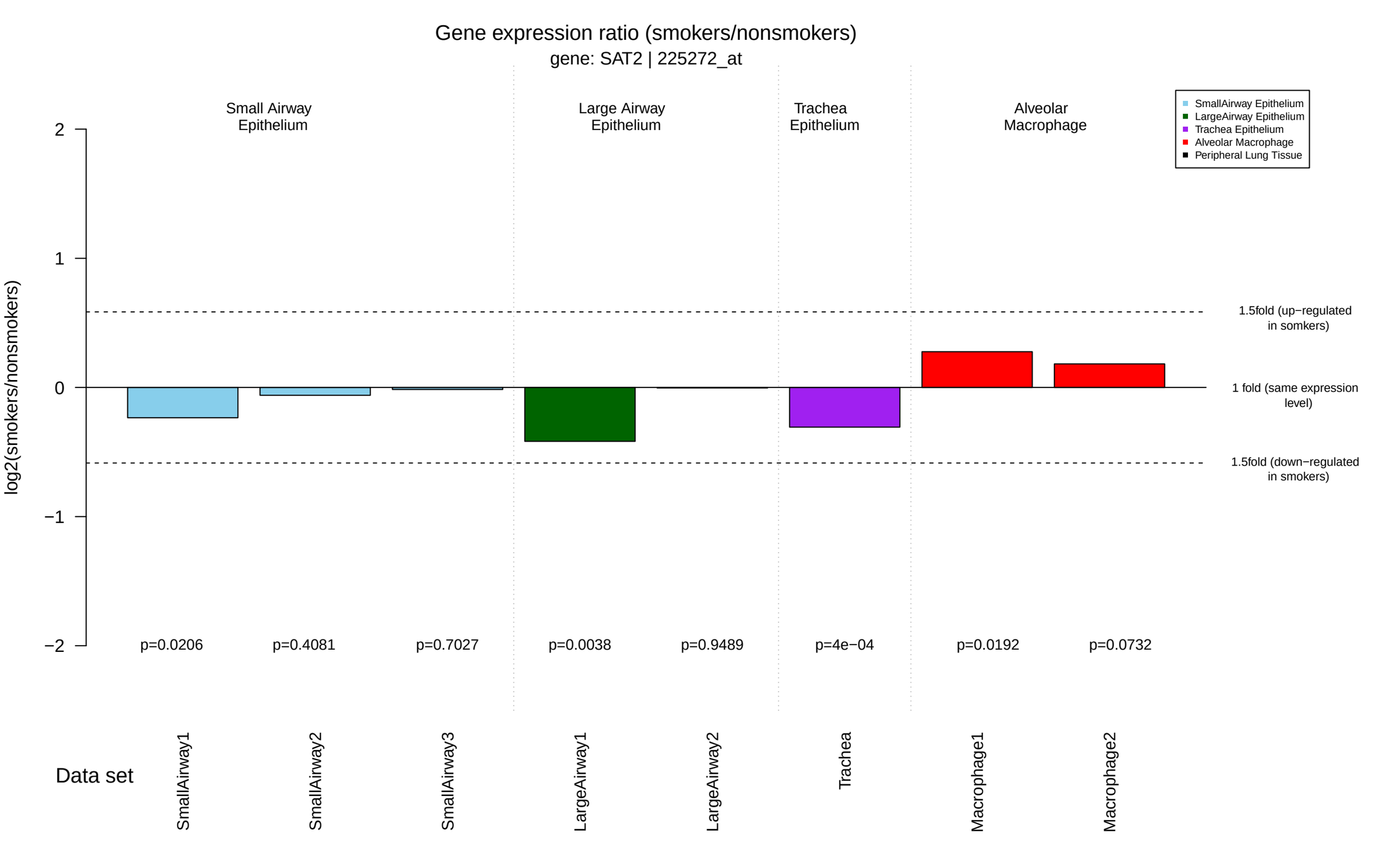
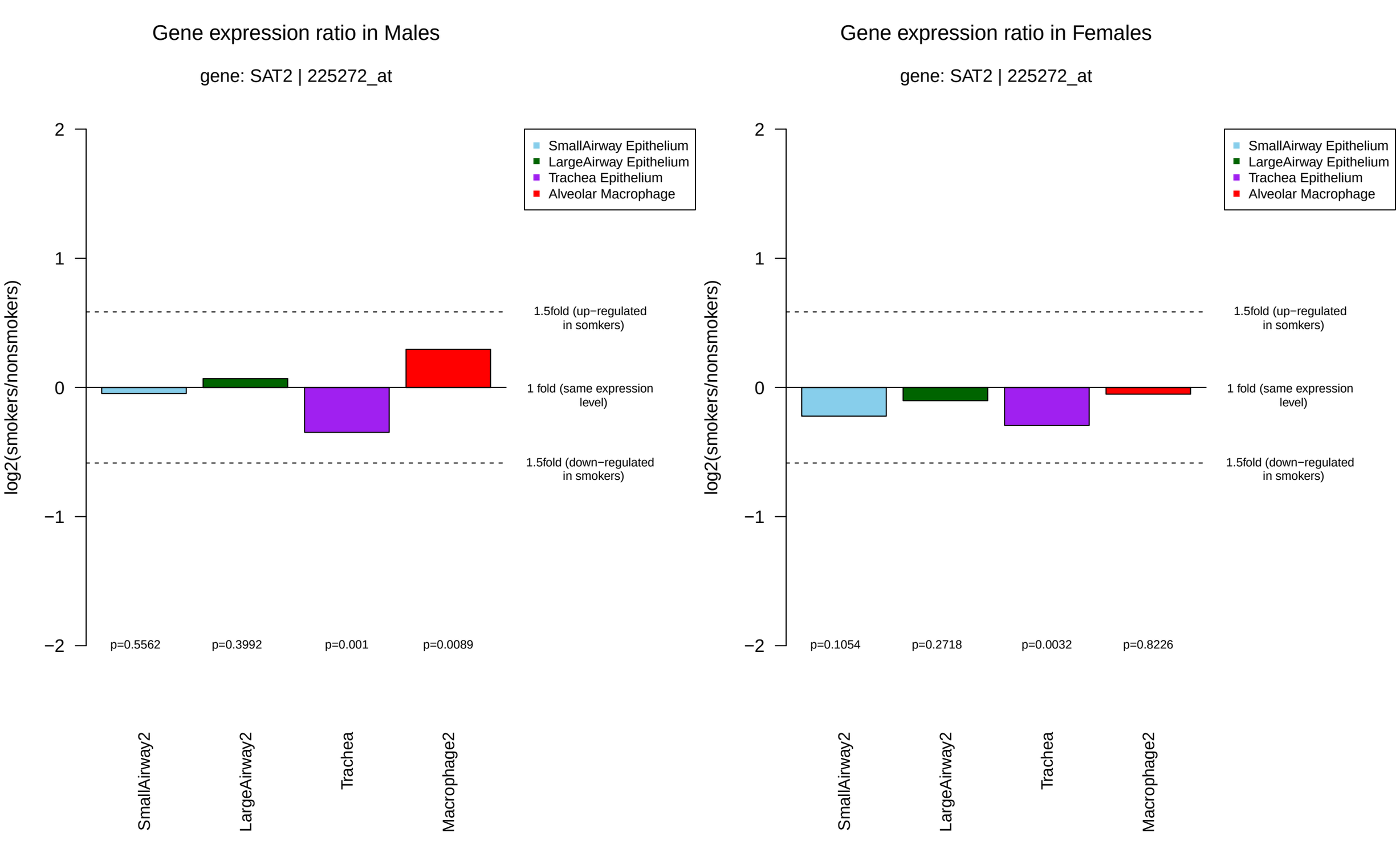
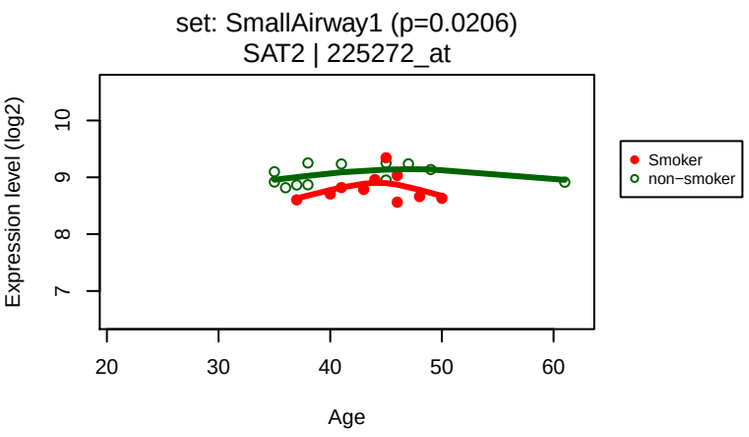
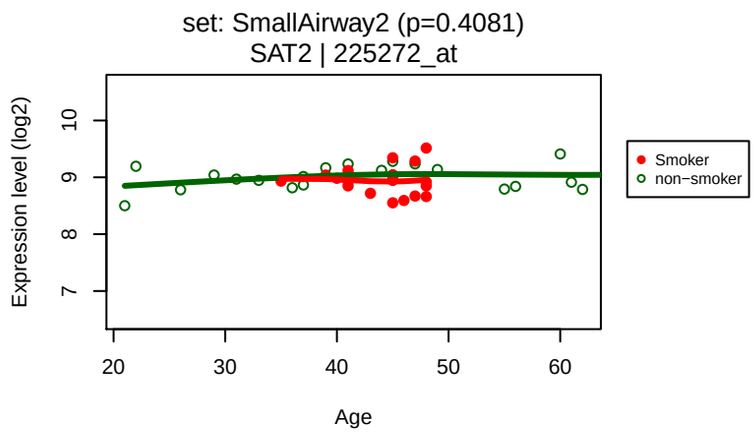
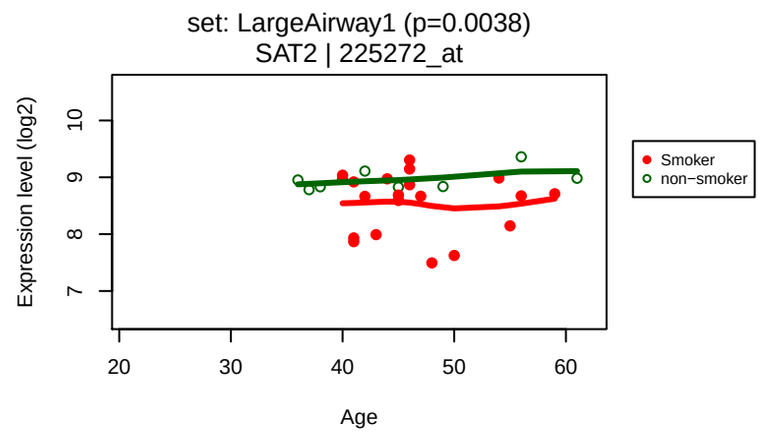

Gene expression of smokers and nonsmokers
gene: SAT2 | 225272_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0206
p=0.4081
p=0.7027
p=0.0038
p=0.9489
p=4e−04
p=0.0192
p=0.0732
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

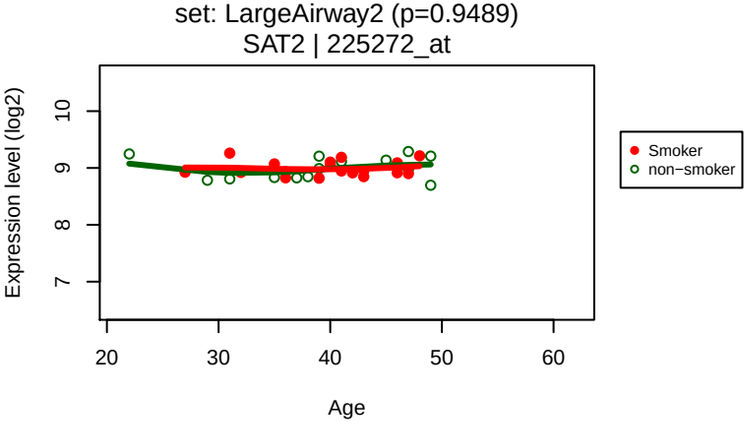

Gene expression ratio (smokers/nonsmokers)
gene: SAT2 | 225272_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0206
p=0.4081
p=0.7027
p=0.0038
p=0.9489
p=4e−04
p=0.0192
p=0.0732
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

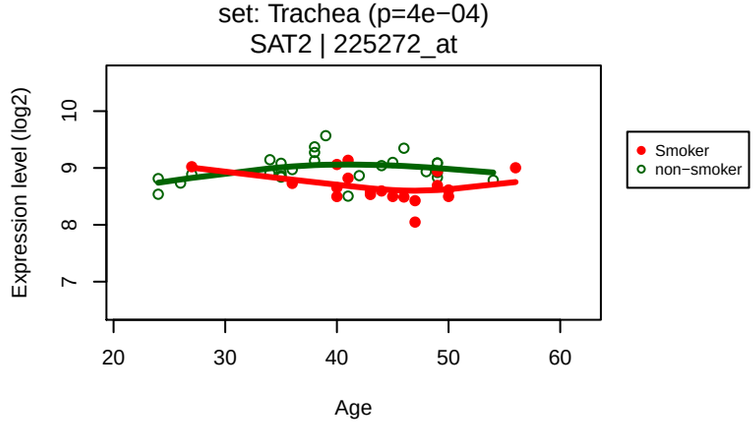

Gene expression ratio in Males
gene: SAT2 | 225272_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5562
p=0.3992
p=0.001
p=0.0089
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SAT2 | 225272_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1054
p=0.2718
p=0.0032
p=0.8226
SmallAirway2
LargeAirway2
Trachea
Macrophage2

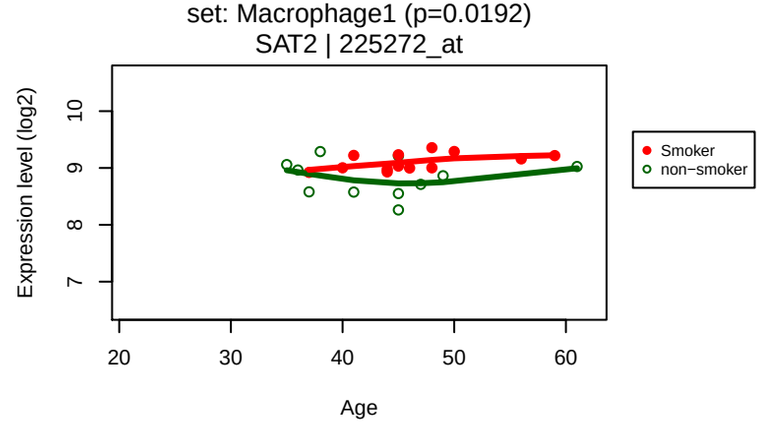

set: SmallAirway1 (p=0.0206)
SAT2 | 225272_at
Expression level (log2)
Age
Smoker
non−smoker

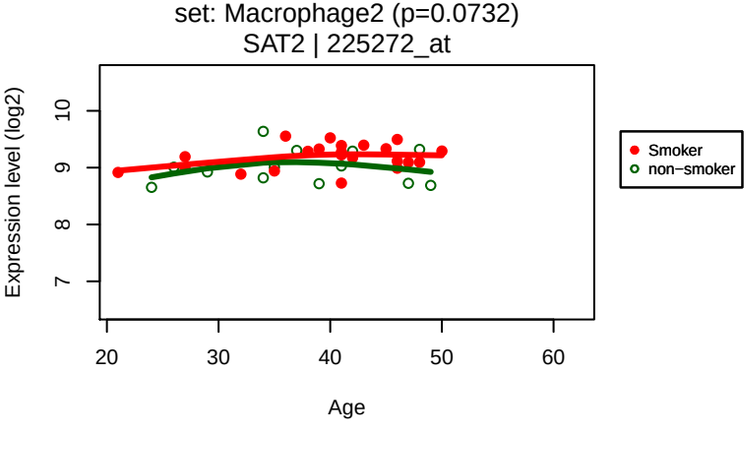

set: SmallAirway2 (p=0.4081)
SAT2 | 225272_at
Expression level (log2)
Age
Smoker
non−smoker

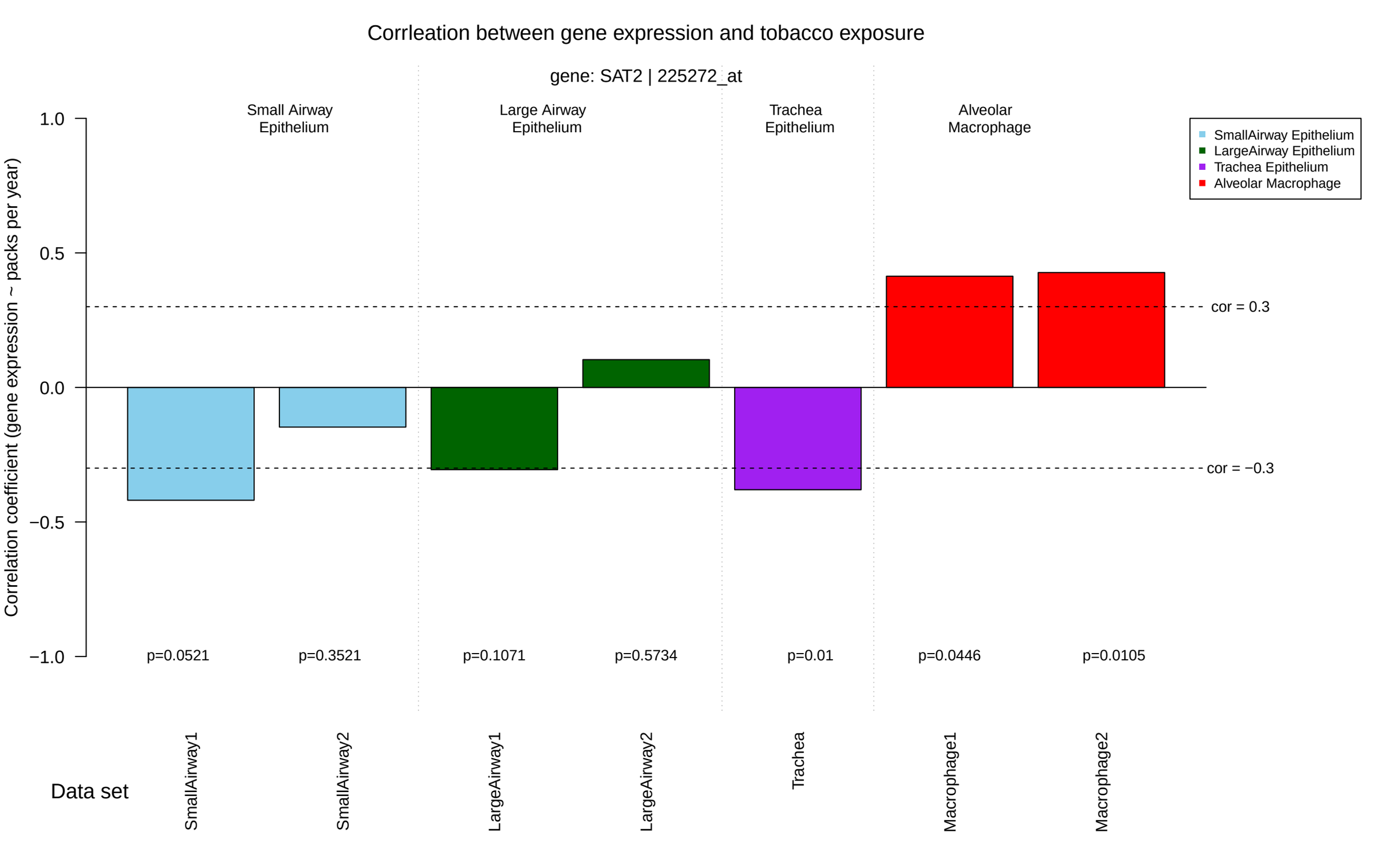

set: LargeAirway1 (p=0.0038)
SAT2 | 225272_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9489)
SAT2 | 225272_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=4e−04)
SAT2 | 225272_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0192)
SAT2 | 225272_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0732)
SAT2 | 225272_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SAT2 | 225272_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0521
p=0.3521
p=0.1071
p=0.5734
p=0.01
p=0.0446
p=0.0105
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2